

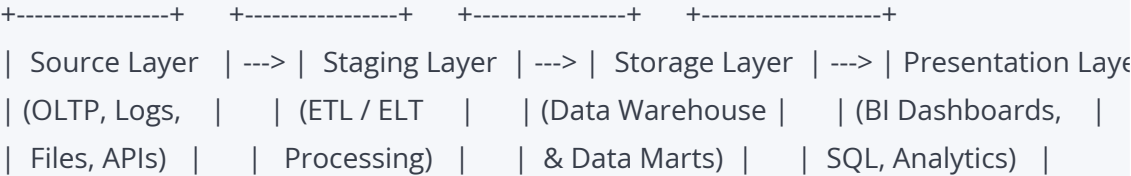


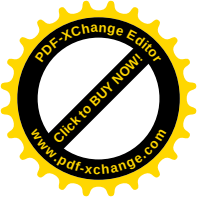
Data Warehousing Summary & Visual Explanations

1. Core Concepts & Architecture

- **Data Warehouse (DW):** A centralized system designed to store and analyze integrated, structured data from multiple sources across an organization to support business intelligence.
- **Data Mart:** A specialized, subject-specific subset of a data warehouse tailored for a single department (e.g., Sales, Finance).
- **Data Lake:** A storage repository holding raw, structured, semi-structured, and unstructured data (documents, media, logs) for flexible future analysis.
- **OLTP vs. OLAP:**
- **OLTP (Online Transaction Processing):** Optimized for fast, frequent, operational write/read operations stored in rows and columns.
- **OLAP (Online Analytical Processing):** Optimized for complex, aggregated, read-only analytical queries using multidimensional structures.

architectural flow:





+-----+ +-----+ +-----+ +-----+

2. Modeling Strategies: Inmon vs. Kimball

Top-Down Approach (Inmon)

[Data Sources]

|
v

+-----+

| Central DW |
| (Normalized) |

+-----+

|
v

+-----+

| Data Marts |
| (Departmental) |

+-----+

Bottom-Up Approach (Kimball)

[Data Sources]

|
v

+-----+

| Data Marts |
| (Denormalized) |

+-----+

|
v

+-----+

| Central DW |
| (Consolidated) |

+-----+

- **Top-Down (Inmon):** Builds a centralized, normalized enterprise data warehouse first, which then populates smaller, departmental data marts.
- **Bottom-Up (Kimball):** Builds dimensional, denormalized data marts for specific business processes first, which are then combined to form the enterprise data warehouse.

3. Dimensional Data Schemas

Star Schema

A central fact table surrounded by and directly joined to individual dimension tables in a single-level hierarchy.

+-----+
| Dim_Customer |



```

      | - Customer_Key (PK) |
      | - Name             |
      | - Region           |
      +-----+
      |
      | (1:N)
+-----+      v      +-----+
| Dim_Date   | ---> +-----+ <--- | Dim_Product   | | |
| - Date_Key (PK) |   | Fact |   | - Product_Key (PK) |
| - Year, Quarter |   | Sales |   | - Category      |
+-----+      +-----+      +-----+

```

Snowflake Schema

An extension of the star schema where dimension tables are normalized into sub-dimensions to reduce redundancy.

```

+-----+      +-----+      +-----+
| Dim_Subcategory | ---> | Dim_Product   | ---> +--+ | Fact_Sales |
| - Subcategory_ID |   | - Product_Key (PK) |   | | +-----+
+-----+      | - Subcategory_ID |   |
                +-----+      |
                    v
+-----+      +-----+      |
| Dim_Country   | ---> | Dim_Customer   | -----+
| - Country_ID  |   | - Customer_Key (PK)|
+-----+      | - Country_ID   |
                +-----+

```

4. Fact & Dimension Table Structure

- **Fact Table:** Contains numerical measurements, metrics, and foreign keys referencing dimension tables (e.g., Sales Amount, Quantity Sold, Unit Cost).
- **Dimension Table:** Contains descriptive metadata and attributes surrounding a business event (e.g., Customer Name, Product Category, Store Location).



FACT TABLE (Sales_Fact)		DIMENSION TABLE (Dim_Product)	
+-----+		+-----+	
Sales_ID (PK)		Product_Key (PK)	
Customer_Key (FK)	----->	Product_Name	
Product_Key (FK)		Category	
Date_Key (FK)		Unit_Price	
Quantity_Sold (Fact)		+-----+	
Total_Amount (Fact)			
+-----+			

5. Data Integration Patterns: ETL vs. ELT

ETL Pattern (Transform before storage):

[Sources] ----> (Extract) ----> [Transform Server] ----> (Load) ----> [Target Warehouse]

ELT Pattern (Transform in cloud target):

[Sources] ----> (Extract) ----> (Load) ----> [Target Cloud Warehouse (Transform)]

- **ETL (Extract, Transform, Load):** Data is transformed on a dedicated processing server before being loaded into the warehouse.
- **ELT (Extract, Load, Transform):** Raw data is loaded directly into the target data warehouse and transformed using the target cloud engine's compute resources.